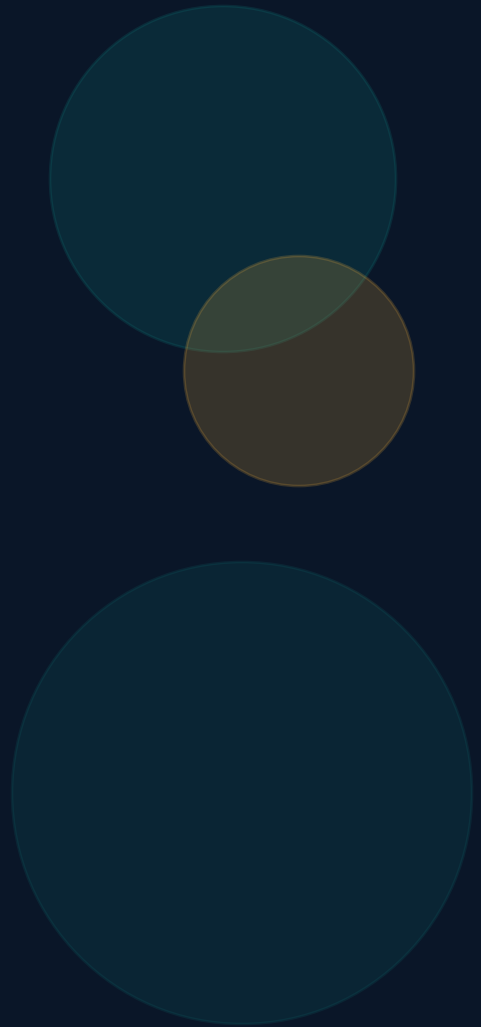


LEARNING ENGAGEMENT Influences Long-Term Knowledge Retention

Statistical Analysis Report | Group Assignment | SLIIT

Descriptive · Inferential · Predictive Analytics



A G E N D A

What We Cover Today

- 01 Research Background & Topic Statement
- 02 Dataset Overview & Variable Selection
- 03 Descriptive Analytics — Engagement & Retention
- 04 Inferential Analytics — Hypothesis Testing
- 05 Predictive Analytics — Modelling Retention
- 06 Key Findings & Conclusion

Research Background & Analytical Statement

Analytical Statement:

"Learning engagement influences long-term knowledge retention"

Research Gap

Traditional measures (attendance, study hours) are assumed to drive learning outcomes — but is this empirically supported?

Dataset Used

Secondary data: Student Performance dataset — 8,000 students, 26 variables across multiple grade levels.

Analytics Used

Three-layer approach: Descriptive → Inferential → Predictive, grounded in statistical modelling principles.

Dataset Overview & Variable Selection

Dataset: **Student Performance & Academic Trends Dataset**

(<https://www.kaggle.com/datasets/ankushnarwade/ai-impact-on-student-performance>)

8,000

Total Students

Balanced sample

26

Raw Variables

Before cleaning

8

Key Variables

Used in analysis

ENGAGEMENT VARIABLES (Independent Variables)

study_hours_per_day

attendance_percentage

study_consistency_index

class_participation_score

tutoring_hours

concept_understanding_score

RETENTION VARIABLES (Dependent Variables)

improvement_rate

final_score

CORE

Study period: One academic semester

OUTCOME

- ✓ All variables are **numeric (continuous/ordinal scales)**
- ✓ Suitable for **correlation analysis**

OUTCOME

OUTCOME

Data Cleaning



Missing Values

Checked selected variables for missing values

No significant missing values after cleaning.



Duplicate Records

Checked dataset for duplicate entries

No duplicate records found.



Variable Types

Verified variables were in correct format for analysis

All variables verified and ready.



Outliers

Examined outliers using boxplots and IQR method

Data distribution is reliable for analysis.



OUTPUT: Cleaned dataset prepared for descriptive analysis

Descriptive Statistics — Engagement & Retention Variables

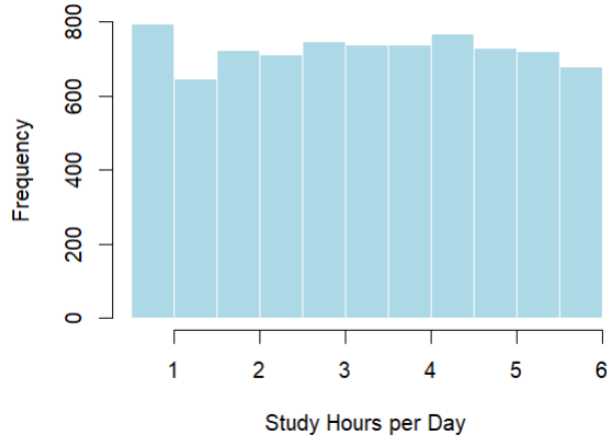
Variable	Mean	Std Dev	Min	Max	Type
study_hours_per_day	3.29	1.58	0.5	6.0	Engagement
attendance_percentage	69.85	17.23	40.0	100.0	Engagement
study_consistency_index	5.52	2.58	1.0	10.0	Engagement
class_participation_score	5.46	2.88	1.0	10.0	Engagement
tutoring_hours	2.52	1.44	0.0	5.0	Engagement
final_score	56.81	13.46	12.7	95.8	Retention
improvement_rate	10.13	17.48	-20.0	40.0	Retention (optional)

Key Observations

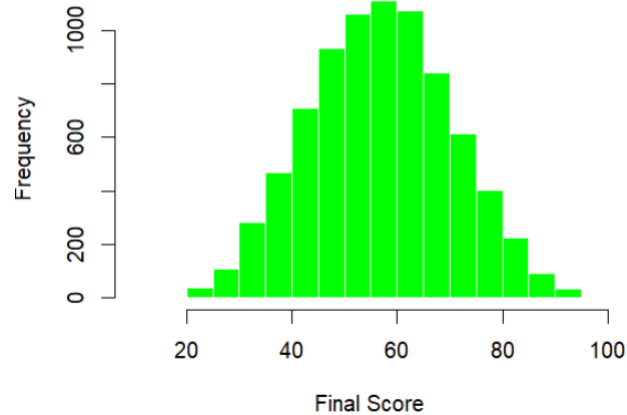
- Engagement variables show moderate variability across students
- Final score exhibits substantial spread (SD ≈ 13.46)
- Improvement rate shows high variability, including negative values
- No obvious extreme skewness observed

Distribution of Engagement and Retention Variables

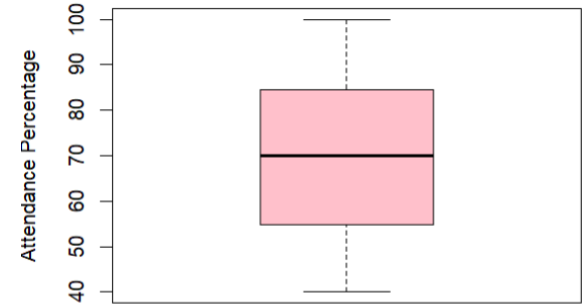
Histogram of Study Hours per Day



Histogram of Final Score



Boxplot of Attendance Percentage

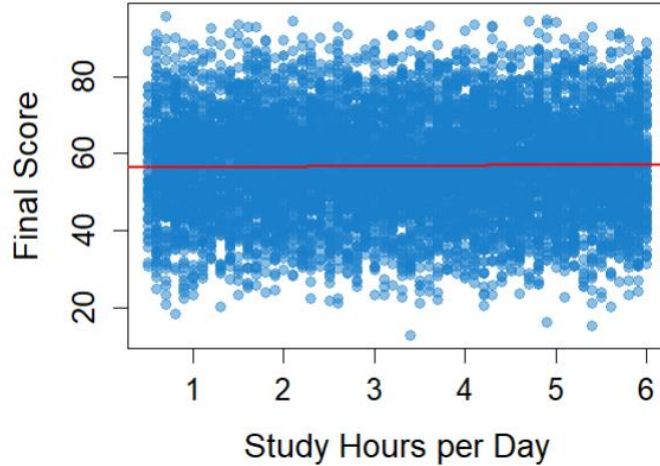


Key Insights

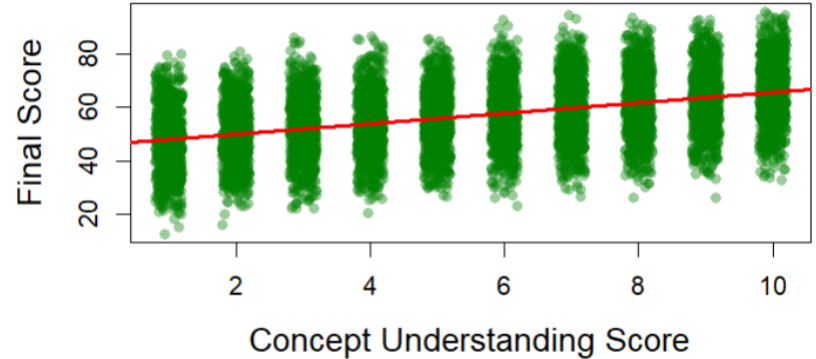
- Study hours are fairly evenly distributed across students
- Final scores approximate a normal distribution
- Attendance shows moderate spread with no extreme outliers

Relationship Between Engagement and Retention Variables

Study Hours per Day vs Final Score



Concept Understanding vs Final Score



Observations

- ✓ No strong linear relationships observed between engagement variables and outcomes
- ✓ Study hours show only a very weak positive trend with final score
- ✓ Attendance shows no clear relationship with improvement rate
- ✓ Visual evidence suggests weak associations → requires statistical testing

Descriptive Summary — Key Patterns in Engagement & Retention



Engagement variables (study hours, attendance, participation) show **moderate variability across students**, indicating diverse study behaviors.



No strong visible relationships between engagement variables and outcomes — **scatter plots show weak or no clear trends**.



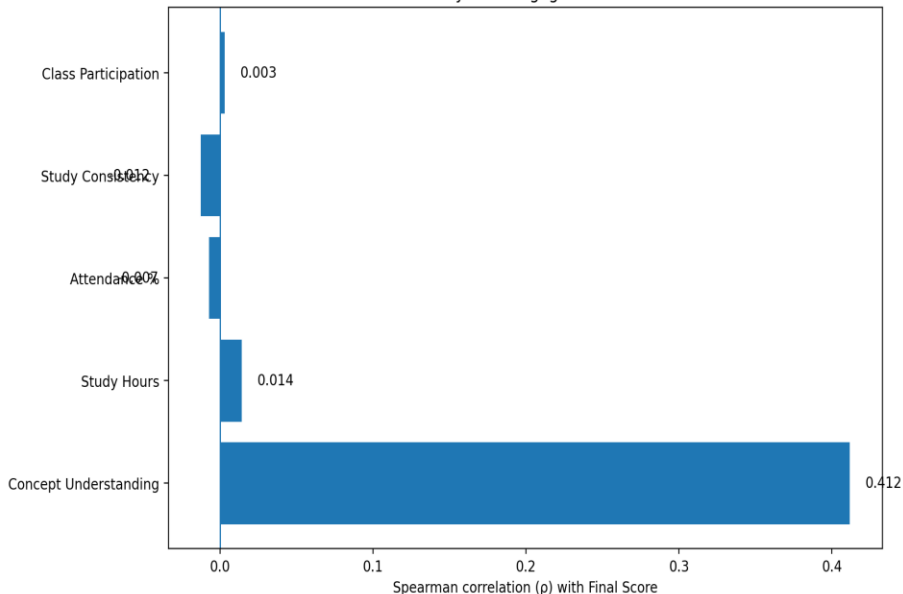
Final score follows an **approximately normal distribution**, while study hours are fairly evenly distributed and attendance shows moderate spread.



Overall distributions and variability suggest the data is **suitable for correlation and inferential analysis**.

Correlation Analysis – Engagement vs Retention

Correlation Analysis — Engagement vs Final Score



Significance Summary			
Variable	Correlation(ρ)	p-value	Significant
Concept Understanding	0.412	<0.001	YES
Study Hours	0.041	0.201	NO
Attendance %	-0.007	0.520	NO
Study Consistency	-0.012	0.264	NO
Class Participation	0.003	0.787	NO

Only concept understanding shows a statistically significant positive relationship with final score ($\rho = 0.412$, $p < 0.001$).
Other engagement variables are not statistically significant predictors

Hypothesis Testing — ANOVA, t-test, and Chi-square

H1: Engagement Group → Final Score (ANOVA)

H_0 : Mean final score is equal across Low, Medium, and High engagement groups.

H_1 : At least one engagement group has a different mean final score.

We reject the null hypothesis since $p < 0.05$.

Engagement level has a significant effect on final score

One-Way ANOVA

$F = 120.2$

$p < 0.0001$

REJECT H_0

H2: Low VS High Engagement → Final Score (t-test)

H_0 : Mean final score is equal for Low and High engagement groups.

H_1 : Mean final score is different for Low and High engagement groups.

We reject the null hypothesis since $p < 0.05$.

High engagement students perform significantly better than low engagement students.

independent Samples t-test

$t = -15.571$

$p < 0.0001$

REJECT H_0

H3: Engagement Level → Pass Status (Chi-square)

H_0 : Engagement and pass status are independent

H_1 : Engagement and pass status are associated

We reject the null hypothesis since $p < 0.05$.

Engagement level is significantly associated with pass status.

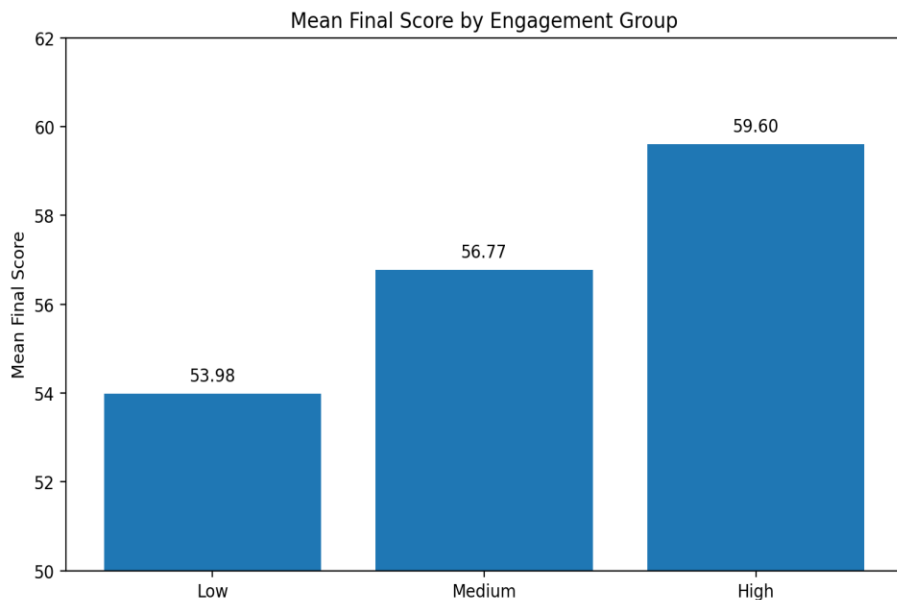
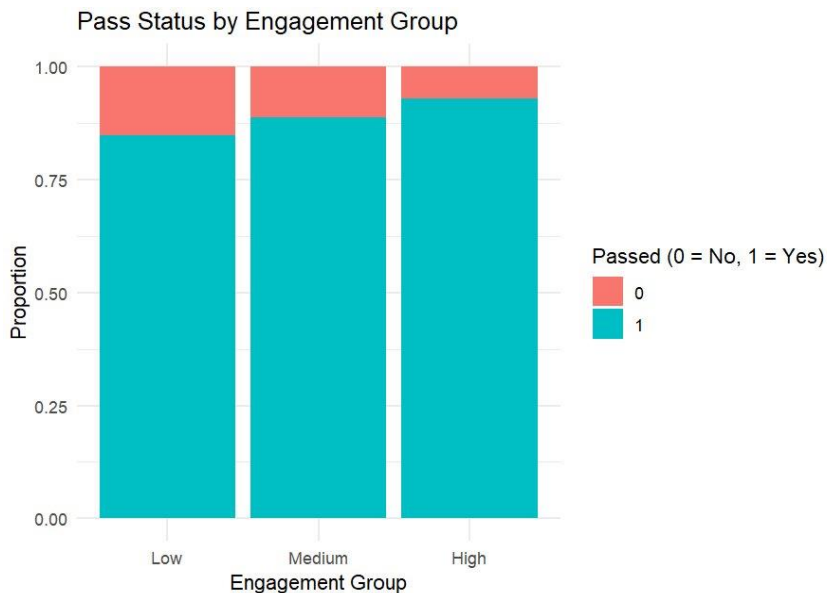
Chi-square Test

$\chi^2 = 86.669$

$p < 2.2e-16$

REJECT H_0

Knowledge Retention Outcomes by Engagement Level



$F = 120.2$

$p < 0.001$

Strong Positive Trend

Multiple Linear Regression — Full Engagement Model

$$\text{final_score} = \beta_0 + \beta_1 \cdot \text{study_hours} + \beta_2 \cdot \text{attendance} + \beta_3 \cdot \text{concept_understanding} + \beta_4 \cdot \text{study_consistency} + \beta_5 \cdot \text{participation} + \varepsilon$$

Predictor	β Estimate	Std. Error	t value	p-value	Significant?
(Intercept)	46.505	0.788	58.998	< 2e-16	—
study_hours_per_day	0.091	0.086	1.055	0.292	✗ No
attendance_percentage	-0.008	0.008	-1.075	0.283	✗ No
concept_understanding_score	1.968	0.047	41.845	< 2e-16 ***	✓ Yes
study_consistency_index	-0.062	0.053	-1.185	0.236	✗ No
class_participation_score	0.028	0.047	0.596	0.551	✗ No

R²

0.1801

18% variation explained

Adj. R²

0.1796

Negligible overfitting

F-statistic

351.1

p < 2.2e-16

Significant predictors

1 of 5

concept_understanding only

Feature Selection — Finding the Best Model

Best Subset Selection

Evaluated all $2^8 = 256$ possible models. Compared using Adjusted R^2 , AIC, BIC, and Mallows C_p .

Result:

concept_understanding_score selected as best 1-variable model

Forward Stepwise

Started with null model, added variables one by one. Stopped when AIC stopped improving.

Result:

concept_understanding_score entered first and last

Backward Stepwise

Started with all 8 variables, removed non-significant ones. Removed 7 variables sequentially.

Result:

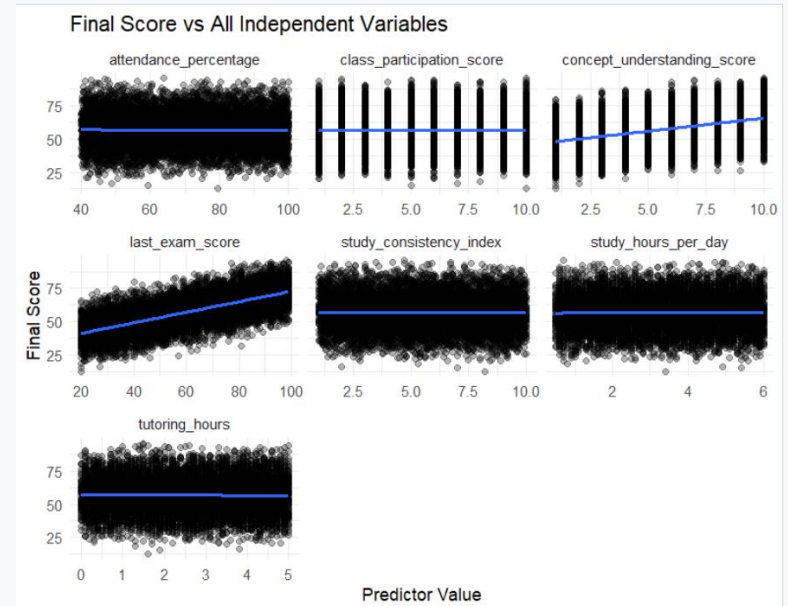
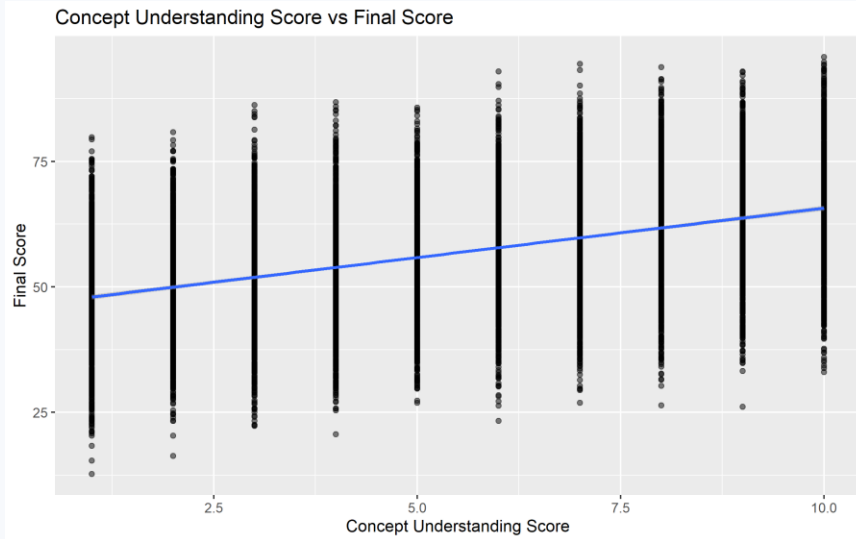
Only concept_understanding_score remained

All three selection methods agree → Final model uses only concept_understanding_score as predictor

Regression Analysis

Multiple Linear Regression Model vs Simple Linear Regression Model

Dependent Variable: Final Score | Predictors: Concept Understanding, Attendance percentage, Study hours, Study consistency, Tutoring hours



MODEL INSIGHT: Simple regression highlights **concept understanding** as the strongest single predictor of final score. Multiple regression confirms that **concept understanding** and **last exam score** are the most influential predictors overall, while the remaining variables contribute less independently.

Final Model — Simple Linear Regression

$$\text{final_score} = 46.02 + 1.97 \times \text{concept_understanding_score} + \varepsilon$$

$\beta_0 = 46.02$ (Intercept)

When concept_understanding_score = 0, the predicted final_score baseline is 46.02

$\beta_1 = 1.97$ (Slope)

Every 1-unit increase in concept understanding $\rightarrow$ +1.97 points in final_score ($p < 2e-16$)

$R^2 = 18\%$ explained

82% unexplained variation (unmeasured factors)

TSS $\sum (y_i - \bar{y})^2$

1,448,271

Total variation in final_score

SSE $\sum (y_i - \hat{y}_i)^2$

1,188,062

Variation model could NOT explain

$R^2 = (\text{TSS} - \text{SSE}) / \text{TSS}$

0.1797

= 260,209 / 1,448,271

Validation (80/20 Train-Test Split): Training $R^2 = 0.1797$ | Test $R^2 = 0.1850$ | RMSE = 12.31 $\rightarrow$ Model generalises well, no overfitting

Partially supported — quality of engagement matters, but quantity does not



concept_understanding_score is the only significant predictor ($\beta = 1.97$, $p < 2e-16$). Cognitive depth of engagement predicts retention.



study_hours, attendance, consistency, participation — all non-significant ($p > 0.05$). Behavioural engagement metrics do not predict final_score.



$R^2 = 0.18$ — engagement explains only 18% of variation. 82% is driven by unmeasured factors: prior knowledge, teaching quality, exam skill, motivation.



Model generalises well — Test R^2 (0.185) $\approx$ Train R^2 (0.180). No overfitting. Methodology is statistically sound.

The low R^2 is not a failure — it is itself a finding that challenges the assumption that simple engagement behaviours drive academic outcomes.

Key Findings & Implications

F1

Descriptive Analytics

Students show moderate engagement and mid-range performance. Data distributions are stable with no major outliers. However, visual analysis indicates weak relationships between engagement and performance, suggesting that further statistical testing is required.

F2

Inferential Analytics

The inferential results provide partial support for the statement that engagement influences knowledge retention. While higher engagement levels are associated with better academic performance, the effect is not consistent across all variables. Concept understanding shows a significant impact, whereas time-related factors do not significantly influence final score.

F3

Consistent Performance Predicts Retention

Multiple regression analysis shows that **concept understanding score is the strongest and most significant engagement predictor of final score ($\beta \approx 2.01$, $p < 0.001$)**.

When academic history is included, **last exam score also becomes a strong predictor ($\beta \approx 0.397$, $p < 0.001$)**.

The model explains up to **65% of the variation in final score ($R^2 = 0.65$)**, indicating strong predictive performance.

However, when considering engagement variables alone, **only concept understanding remains significant**, supporting the idea that **quality of learning matters more than quantity**.

Learning Engagement & Long-Term Knowledge Retention

The analytical statement is JUSTIFIED:

Our results show that engagement does impact retention, but not all engagement factors are important. Concept understanding plays the biggest role, while time-based variables are not significant. This means students need to focus more on understanding rather than just spending more time studying. Also, since the model doesn't explain everything, other factors also affect performance

Methods Used:

Descriptive Analytics · Pearson Correlation · One-Way ANOVA · Multiple Linear Regression | Dataset: n = 8,000